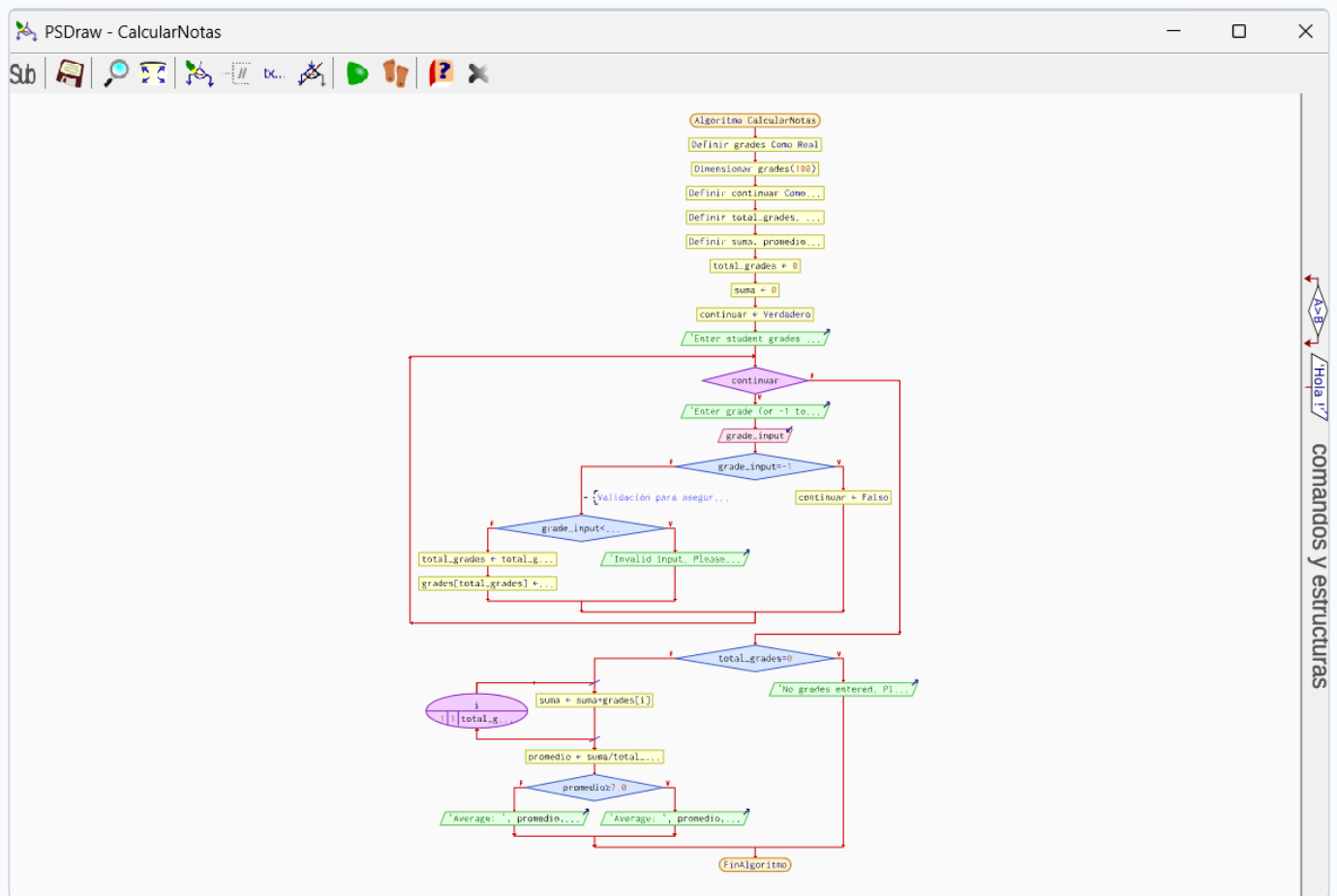


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HW 2 - U2 - Selective and Iterative Control Structures

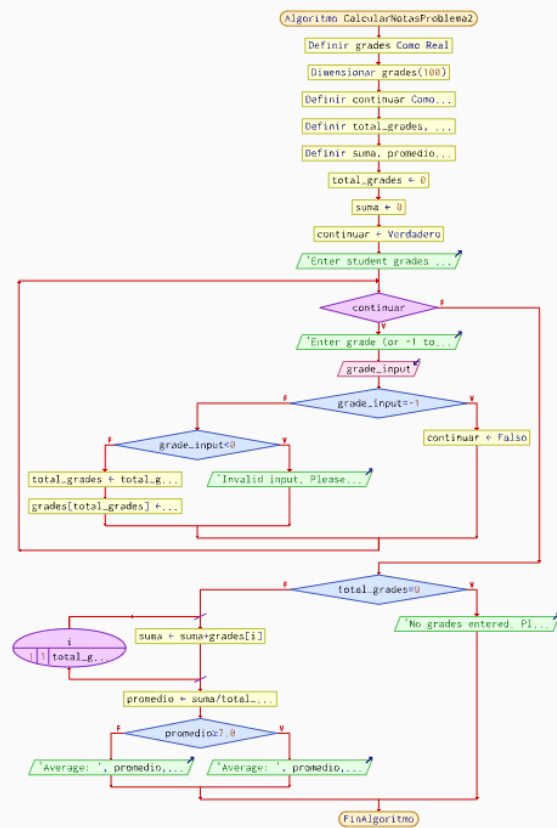
Código del Problema 1

```
#Problem 1:
def calculate_grades():
    grades= []
    print("Enter student grades one by one. Type 'done' when finished.")
    while True:
        user_input = input("Enter grade (or 'done'): ").strip().lower()
        if user_input == 'done':
            break
        try:
            grade = float(user_input)
            grades.append(grade)
        except ValueError:
            print("Invalid input. Please enter a valid number or 'done'.")
    if not grades:
        print("No grades entered. Please enter at least one grade.")
        return
    average = sum(grades) / len(grades)
    status = "Passed" if average >= 7.0 else "Failed"
    print(f"Average: {average:.2f} – {status}")
```



Código del Problema 2

```
#Problem 2:
def calculate_grades():
    grades= []
    print("Enter student grades one by one. Type 'done' when finished.")
    while True:
        user_input = input("Enter grade (or 'done'): ").strip().lower()
        if user_input == 'done':
            break
        try:
            grade = float(user_input)
            grades.append(grade)
        except ValueError:
            print("Invalid input. Please enter a valid number or 'done'.")
    if not grades:
        print("No grades entered. Please enter at least one grade.")
        return
    average = sum(grades) / len(grades)
    status = "Passed" if average >= 7.0 else "Failed"
    print(f"Average: {average:.2f} - {status}")
calculate_grades()
```



Código del Problema 3

```
#Problem 3
def water_bill_calculator():
    print("Water Bill Calculator – Enter monthly consumption. Type 'exit' to stop.")

    total_accumulated_charge = 0.0

    while True:
        user_input = input("\nEnter m³ consumed (or type 'exit'): ").strip().lower()

        if user_input == 'exit':
            break

        try:
            m3 = float(user_input)

            if m3 < 0:
                print("Consumption cannot be negative.")
                continue

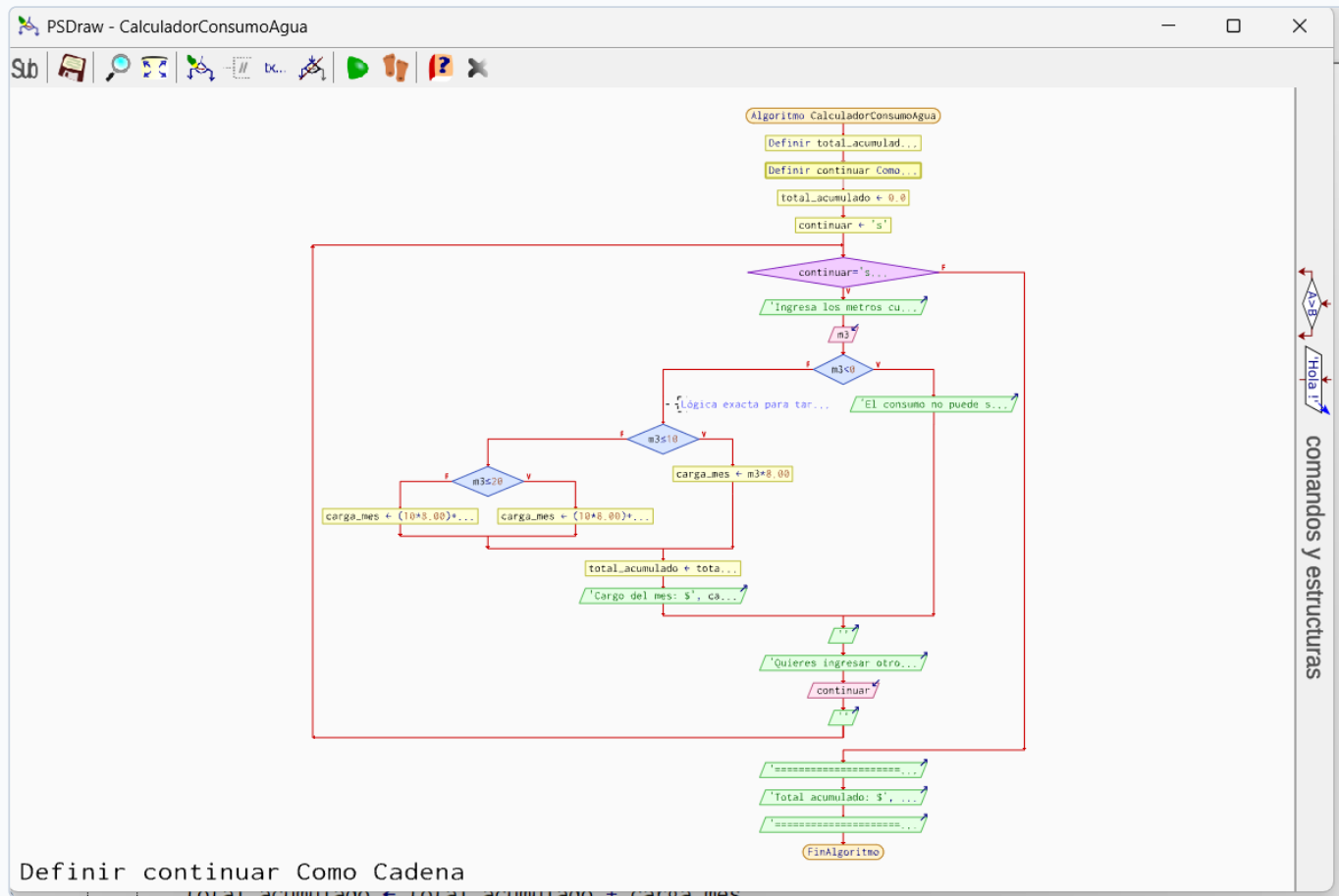
            if m3 <= 10:
                rate = 8.00
            elif m3 <= 20:
                rate = 12.00
            else:
                rate = 18.00

            month_charge = m3 * rate
            total_accumulated_charge += month_charge

            print(f"Month charge: ${month_charge:.2f} MXN")

        except ValueError:
            print("Invalid input. Please enter a valid number or 'exit'.")

    print(f"Total: ${total_accumulated_charge:.2f} MXN")
water_bill_calculator()
```



Código del Problema 4

```
#Problem 4
def season_classifier():
    print("Season Classifier – Enter month numbers (1-12). Type 'exit' to stop.")

    while True:
        user_input = input("\nEnter month number (or type 'exit'): ").strip().lower()

        if user_input == 'exit':
            print("Exiting program.")
            break

        try:
            month = int(user_input)

            if month < 1 or month > 12:
                print("Invalid month. Please enter a number between 1 and 12.")
                continue

            if month in [12, 1, 2]:
                season = "Winter"
            elif month in [3, 4, 5]:
                season = "Spring"
            elif month in [6, 7, 8]:
                season = "Summer"
            else:
                season = "Fall"

            print(season)

        except ValueError:
            print("Invalid input. Please enter a valid number or 'exit'.")
    season_classifier()
```



```

#problem 5
def shipping_cost_calculator():
    print("Shipping Cost Calculator – Enter package details. Type 'exit' to stop.")

    total_accumulated_cost = 0.0

    while True:

        weight_input = input("\nEnter weight in kg (or type 'exit'):"
        ).strip().lower()

        if weight_input == 'exit':
            break

        try:
            weight = float(weight_input)

            distance_input = input("Enter distance in km: ").strip()
            distance = float(distance_input)

            if weight <= 0 or distance <= 0:
                print("Weight and distance must be greater than 0.")
                continue

            if distance <= 100:
                if weight <= 5:
                    cost = 50.00
                else:
                    cost = 80.00
            else:
                if weight <= 5:
                    cost = 120.00
                else:
                    cost = 200.00

            total_accumulated_cost += cost
            print(f"Shipping cost: ${cost:.2f} MXN")

        except ValueError:
            print("Invalid input. Please enter numbers or type 'exit'.")

    print(f"Total: ${total_accumulated_cost:.2f} MXN")

shipping_cost_calculator()

```

